

Lecture 3: Calculus and Linear Algebra

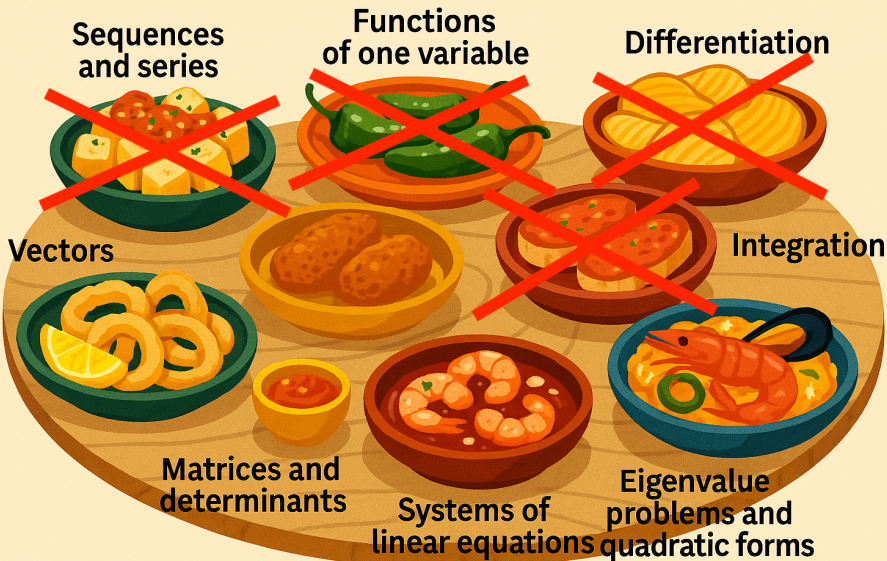
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Course content (math tapas)



Orthogonality and linear independence

Orthogonality

Definition: $\mathbf{u} \perp \mathbf{v}$ if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Angle example: For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (5, -4, -1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 11, \quad \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{11}{\sqrt{14}\sqrt{42}} \approx 0.4537,$$

so the angle is about 63° .

Geometric test: $\mathbf{u} \perp \mathbf{v}$ if and only if $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Cauchy-Schwarz via completing the square

Assume nonzero $\mathbf{u}, \mathbf{v}$. For any real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

Complete the square in t :

$$\|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 = \|\mathbf{v}\|^2 \left(t - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \right)^2 + \left(\|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2} \right).$$

Since the left side is ≥ 0 for all t , the second term must be ≥ 0 :

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

Equality condition: equality holds if and only if $\|\mathbf{u} - t\mathbf{v}\|^2 = 0$ for $t = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, i.e., $\mathbf{u} = t\mathbf{v}$ (collinear vectors).



Linear dependence, independence, and bases

Linear combination: $\mathbf{u} = \sum_{i=1}^m \lambda_i \mathbf{v}_i$.

Linear independence: $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_m = 0.$$

Basis: Any n linearly independent vectors in $\mathbb{R}^n$ form a basis. Then every $\mathbf{u} \in \mathbb{R}^n$ has a **unique** representation $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$.

Standard basis: $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Say $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_3$. Show that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are NOT linearly independent.

Orthogonal complements and other subspaces

Subspaces, span, and dimension

For $E = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, the **span** is

$$\text{span}(E) = \left\{ \sum_{i=1}^k \lambda_i \mathbf{v}_i : \lambda_i \in \mathbb{R} \right\}.$$

A **subspace** $V \subset \mathbb{R}^n$ is any span. A **basis of V** is a linearly independent set that spans V . The **dimension** of V is the size of any basis.

Exercise: Basis of $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Orthogonal complement and projection

Orthogonal complement

For a subspace $V \subset \mathbb{R}^n$,

$$V^\perp = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0 \ \forall \mathbf{v} \in V\}.$$

Projection theorem

For any $\mathbf{y} \in \mathbb{R}^n$ and subspace V , there is a unique $\hat{\mathbf{y}} \in V$ with $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. We call $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto V .

Least squares view: With design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and response $\mathbf{y}$, the LS fit $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Suppose $n = 2$ and $V = \text{span}\{\mathbf{u}\}$. Take any $\mathbf{y} \in \mathbb{R}^2$.

We have

$$V^\perp = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0 \ \forall \mathbf{v} \in V\} = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{u} \rangle = 0\}.$$

The projection $\mathbf{y}$ must satisfy

- $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$ or, equivalently, $\langle \mathbf{y}, \mathbf{u} \rangle = \langle \hat{\mathbf{y}}, \mathbf{u} \rangle$.
- $\hat{\mathbf{y}} \in V$ or, equivalently, $\hat{\mathbf{y}} = \lambda \mathbf{u}$.

Using these two properties, we get $\hat{\mathbf{y}} = \lambda \mathbf{u}$, where

$$\lambda = \frac{\langle \mathbf{y}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle}$$



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Chapter 6: Matrices and determinants

A **matrix** $A \in \mathbb{R}^{m \times n}$ is a table with m rows, n columns. The (i, j) entry is a_{ij} .

We recall basic facts about matrices.

Why care: matrices express linear maps, data tables, network flows, input-output models, regressions, transformations, and more.

Read Werner-Sotskov Ch. 7; Simon-Blume Chs. 8-9.

Exercises 7.6, 7.9(b,c,d), 7.12, 7.14(a), 7.16, 7.18

Matrix basics

Matrices

A table of numbers with m rows and n columns: $A \in \mathbb{R}^{m \times n}$. The (i, j) -th entry is denoted by a_{ij} .

Special matrices:

- zero matrix: $\mathbf{0}_{m \times n} \in \mathbb{R}^{m \times n}$
- identity matrix: $\mathbb{I}_n \in \mathbb{R}^{n \times n}$.
- **transposition**: $A^T \in \mathbb{R}^{n \times m}$, $(A^T)_{ij} = a_{ji}$.
- square matrix: $m = n$.
- symmetric matrix: square matrix such that $A = A^T$.
- diagonal matrix: $a_{ij} = 0$ if $i \neq j$.
- lower (upper) triangular: $a_{ij} = 0$ if $i < j$ ($i > j$).
- $\mathbf{v} \in \mathbb{R}^n$ is treated as $n \times 1$ matrix, $\mathbb{R}^n \simeq \mathbb{R}^{n \times 1}$.

Basic matrix operations

Addition and scalar multiplication are entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

They satisfy the usual commutative, associative, and distributive laws.

Matrix product: If $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$, then $C = AB \in \mathbb{R}^{m \times n}$ with

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

Example:

$$\begin{pmatrix} 2 & 3 & 4 & 1 \\ 7 & -1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 7 \\ 3 & -1 \\ 4 & 0 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 30 & 15 \\ 15 & 66 \end{pmatrix}.$$

Algebraic properties

Facts:

1. $(AB)C = A(BC)$
2. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
3. Generally **not** commutative: $AB \neq BA$
4. $AI_n = I_m A = A$ (dimensions must match)
5. $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$, $(AB)^T = B^T A^T$

Remark: AA^T is always symmetric.

Matrix times vector

We treat $\mathbb{R}^n$ as column vectors $\mathbb{R}^{n \times 1}$. If $A \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$ then $A\mathbf{x} \in \mathbb{R}^m$ is a linear combination of the columns of A with coefficients x_i :

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

e.g. **Another look at the LS method:** $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$

$$\text{minimize}_{\beta \in \mathbb{R}^d} \quad \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Finds the closest point to $\mathbf{y}$ in the space spanned by the columns of $\mathbf{X}$.

Orthogonal projection

Theorem: Given a vector $\mathbf{y} \in \mathbb{R}^n$ and a subspace $V \subset \mathbb{R}^n$ there exists a unique $\hat{\mathbf{y}} \in V$ such that $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. Let $\mathbf{x}_1, \dots, \mathbf{x}_d$ be a basis of $V \subset \mathbb{R}^n$.

$\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_d$.

$\hat{\mathbf{y}} \in V$ means $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\lambda}$ for some $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$.

$\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$ means $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$.

The unique solution: $\boldsymbol{\lambda} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ giving

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

(We still need to show that $\mathbf{X}^\top \mathbf{X}$ is invertible.)

Matrix inverse

A square matrix A is **invertible** if there exists a matrix A^{-1} such that

$$AA^{-1} = I_n.$$

We want to show:

$$A \in \mathbb{R}^{n \times n} \text{ is invertible} \iff A\mathbf{x} = \mathbf{0}_n \text{ only for } \mathbf{x} = \mathbf{0}_n.$$

Two observations:

1. $A\mathbf{x} = \mathbf{0}_n$ only for $\mathbf{x} = \mathbf{0}_n$ if and only if columns of A are linearly independent.
2. n independent vectors in $\mathbb{R}^n$ form a basis and so $\forall i = 1, \dots, n \exists \mathbf{b}_i$ such that $A\mathbf{b}_i = \mathbf{e}_i$.
3. This gives $B \in \mathbb{R}^{n \times n}$ such that $AB = \mathbb{I}_n$.

This is enough to show that the matrix $\mathbf{X}^\top \mathbf{X}$ on slide 15 is invertible.

Note: If $(\mathbf{X}^\top \mathbf{X})\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})\mathbf{x} = 0$ but this only possible if $\mathbf{x} = \mathbf{0}$.

Side remark

We defined the inverse of A as the matrix A^{-1} such that $AA^{-1} = I_n$.

But of course, equivalently, $A^{-1}A = I_n$.

Indeed, if $AA^{-1} = I_n$ then $AA^{-1}A = A$ or, equivalently,

$$A(A^{-1}A - I_n) = \mathbf{0}_{n \times n}.$$

Using the fact that the columns of $A\mathbf{x} = \mathbf{0}_n$ if and only if $\mathbf{x} = \mathbf{0}_n$, we can conclude that each column of $A^{-1}A - I_n$ is zero, and so

$$A^{-1}A - I_n = \mathbf{0}$$

as claimed.

Matrix subspaces and decompositions

Important spaces and rank

For $A \in \mathbb{R}^{m \times n}$:

- **Column space** (image) $\text{Im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subset \mathbb{R}^m$
- **Row space** $\text{Im}(A^T) \subset \mathbb{R}^n$
- **Kernel** (null space) $\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \subset \mathbb{R}^n$
- **Rank** $\text{rank}(A) = \dim \text{Im}(A) = \dim \text{Im}(A^T)$

Orthogonality: Row space is orthogonal to $\ker(A)$.

Rank-nullity: $\text{rank}(A) + \dim \ker(A) = n$.

Try: $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15



1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\mathbf{1}_4 = \mathbf{33} \cdot \mathbf{1}_4 \quad \text{and}$$

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\mathbf{1}_4^\top = \mathbf{33} \cdot \mathbf{1}_4^\top.$$

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\text{trace}(\mathbf{A}) = \mathbf{33},$$

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\det(\mathbf{A}) = -1485 = 45 \cdot \mathbf{33}.$$

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\text{rank}(\mathbf{A}) = 4$$

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15

$$\ker(\mathbf{A}) = \{\mathbf{0}_4\}.$$

Row/column space are both equal to $\mathbb{R}^4$

Orthogonal matrices

An $n \times n$ matrix A is called **orthogonal** if $AA^\top = I_n$.

Remark: If A is orthogonal, its row vectors (and also its column vectors) are pairwise orthogonal unit vectors.

Proof: Let $\mathbf{r}_i$ and $\mathbf{r}_j$ be the i -th and j -th rows of A . The (i, j) entry of $AA^\top$ is the scalar product $\mathbf{r}_i^\top \mathbf{r}_j$. If $AA^\top = I$, then $\mathbf{r}_i^\top \mathbf{r}_j = 0$ for $i \neq j$ (orthogonality) and $\mathbf{r}_i^\top \mathbf{r}_i = \|\mathbf{r}_i\|^2 = 1$ (unit length). The same holds for columns using $A^\top A = I_n$.

Example:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Singular value decomposition (SVD)

Singular value decomposition

For every $A \in \mathbb{R}^{m \times n}$ there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and a “diagonal” matrix $\Lambda \in \mathbb{R}^{m \times n}$ such that

$$A = U\Lambda V^T.$$

If $m \leq n$ then Λ has m “diagonal” entries.

We can assume $\Lambda_{11} \geq \dots \geq \Lambda_{mm} \geq 0$.

These diagonal entries are called **singular values** of A .

Spectral decomposition

Singular value decomposition

Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric. Then there exist orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal Λ such that

$$A = U\Lambda U^{\top}.$$

Denote $(\lambda_1, \dots, \lambda_n) = \text{diag}(\Lambda)$.

We take $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$.

Column/row reductions and determinant

Elementary matrix operations

Column or row operations:

1. **Swap** two columns (or rows)
2. **Scale** a column (or row) by $\lambda \neq 0$
3. **Add** a multiple of one column (or row) to another

Each is implemented by multiplying by a suitable elementary matrix on the right (for column ops) or left (for row ops). Useful for Gaussian elimination and determinant computation.

To see what is happening, let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and multiply from the right by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}$, or $\begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}$.

Determinants: definition

For $A \in \mathbb{R}^{n \times n}$, let A_{ij} be the submatrix with row i and column j removed. Define recursively

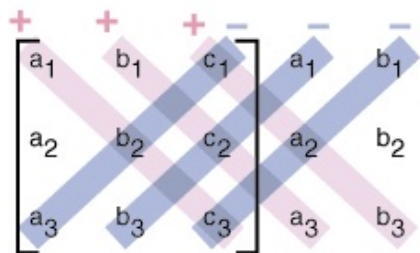
$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}), \quad \det([a_{11}]) = a_{11}.$$

For $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

For $n = 3$:

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

A memory aid for $n = 3$



$$\det A = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1)$$

Cofactor expansion and properties

Cofactor expansion: expand by any row i or any column j :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Properties:

1. $\det(A) = \det(A^T)$
2. If A is triangular, $\det(A)$ is the product of diagonal entries
3. $\det(AB) = \det(A) \det(B)$
4. Swapping two rows (or columns) flips the sign of $\det$
5. Scaling a row (or column) by λ scales $\det$ by λ
6. Adding a multiple of one row to another leaves determinant unchanged
7. $\det(A) = 0$ if and only if rows (or columns) are linearly dependent

Determinants by elimination

Use row operations (keeping track of determinant changes) to reach triangular form.

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & 3 \end{vmatrix} = (-6) \cdot 3 = -18.$$

First: $r_2 \leftarrow r_2 - 3r_1$. Then: $r_3 \leftarrow r_3 - r_1$.

Geometric meaning: $|\det(A)|$ is the area/volume scaling of the linear map $x \mapsto Ax$ (and the sign encodes orientation).

2D example:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det(A) = 1.$$

The unit square is mapped to a parallelogram of the same area but tilted. If $\det(A) = -1$, the square would be flipped (orientation reversed).

Linear mappings and their inverse

Linear mappings

A mapping $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A(\mathbf{x}_1) + A(\mathbf{x}_2), \quad A(\lambda \mathbf{x}) = \lambda A(\mathbf{x}).$$

Then there exists an $m \times n$ matrix (also denoted A) with $A(\mathbf{x}) = A\mathbf{x}$.

Columns as images: $A(\mathbf{e}_i) = \mathbf{a}_i$ (the i -th column), and $A(\mathbf{x}) = \sum_i x_i A(\mathbf{e}_i)$.

Examples: scalings, rotations, reflections, projections, feature maps in ML, Leontief input-output in economics.

Two simple linear maps

1. Reflection across the y -axis:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}.$$

2. Rotation by 45° counterclockwise:

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Check the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ to see the action.

Inverse matrix and properties

A square A is invertible if A^{-1} exists with $AA^{-1} = A^{-1}A = I_n$.

Properties:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(\lambda A)^{-1} = \lambda^{-1}A^{-1}$ for $\lambda \neq 0$
5. $\det(A^{-1}) = 1/\det(A)$

Solve $A\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = A^{-1}\mathbf{b}$ when A is invertible.

Computing inverses

Cofactor formula: If A is nonsingular,

$$A^{-1} = \frac{1}{\det A} C(A)^T, \quad C(A)_{ij} = (-1)^{i+j} \det(A_{ij}).$$

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For larger n , numerical methods use elimination (LU), not cofactors.

Example: Input-output model

Let a_{ij} be units of good i needed to produce 1 unit of good j . Put $A = (a_{ij})$. Let $\mathbf{x}$ be total output and $\mathbf{y}$ the vector of final demand.

Accounting identity: output = internal demand + final demand

$$\mathbf{x} = A\mathbf{x} + \mathbf{y} \quad \Leftrightarrow \quad (I_n - A)\mathbf{x} = \mathbf{y} \quad \Rightarrow \quad \mathbf{x} = (I_n - A)^{-1}\mathbf{y}$$

provided $I_n - A$ is invertible.

Interpretation: $(I - A)^{-1} = I + A + A^2 + \dots$ accumulates direct, indirect, and higher-order input needs when it converges.

A triangular example

Theorem: If A is strictly upper triangular (zeros on and below diagonal), then $A^n = 0$ and

$$(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{n-1}.$$

Check $(I - A)(I + A + \cdots + A^{n-1}) = I - A^n = I$.

Example:

$$A = \begin{pmatrix} 0 & 3 & 5 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = (66, 15, 4).$$

Chapter 7: Systems of Linear Equations

Many problems in economics and data science can be modeled as a **system of linear equations**.

In this chapter we recall core facts and **general solution procedures**.

Read Chapter 8 of Werner–Sotskov; Chapter 7 of Simon–Blume.

Exercises 8.5, 8.6, 8.8 (Werner–Sotskov).

System of linear equations

Systems of linear equations

Let A be an $m \times n$ matrix, $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ unknown, and $\mathbf{b} \in \mathbb{R}^m$ given. The system

$$A\mathbf{x} = \mathbf{b}$$

has the **vector form**

$$\sum_{j=1}^n x_j \mathbf{a}_j = \mathbf{b},$$

where $\mathbf{a}_1, \dots, \mathbf{a}_n$ are the columns of A .

If $\mathbf{b} = \mathbf{0}$ the system is **homogeneous**; otherwise it is **inhomogeneous**.

An inhomogeneous system can be **inconsistent** (no solution) or **consistent** (at least one solution).

A homogeneous system always has $\mathbf{x} = \mathbf{0}$ as a solution.

Motivating example: ranking the web (PageRank)

Modern search engines rank pages by modeling a **random surfer**: a user who clicks links from page to page.

Let P be the **transition matrix** where P_{ij} is the probability of moving from page i to j . In the simplest model, the long-run probabilities $\mathbf{x}$ satisfy

$$\mathbf{x} = P^\top \mathbf{x},$$

so $\mathbf{x}$ is the stationary distribution of the Markov chain.

Issue: If a page has no outgoing links, or if the web splits into parts, this system can break down (no unique stationary distribution).

Teleportation and damping

To fix these problems, PageRank introduces two ingredients:

- **Teleportation vector $\mathbf{v}$:** with some probability, the surfer jumps to any page instead of following links. Often chosen uniform: $\mathbf{v} = \frac{1}{n}(1, \dots, 1)$ so every page is reachable.
- **Damping factor α :** probability of *following links*. Typical choice: $\alpha = 0.85$ (85% follow a link, 15% teleport).

The modified PageRank system is:

$$(I - \alpha P^T) \mathbf{x} = (1 - \alpha) \mathbf{v}.$$

Result: This ensures a unique ranking vector $\mathbf{x}$ for any web structure.

Computing the rank with Gaussian elimination

Elementary row or column operations do not change $\text{rank}(A)$.

$$\begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & -4 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

The **pivot** in each nonzero row is the first nonzero entry from the left.

There are 2 pivots, hence $\text{rank}(A) = 2$.

To find a linear relation among columns, solve the homogeneous system for $(\lambda_1, \lambda_2, \lambda_3)$:

$$\lambda_1 + 2\lambda_2 = 0, \quad -\lambda_2 + \lambda_3 = 0.$$

Thus $(\lambda_1, \lambda_2, \lambda_3) = (-2\lambda_3, \lambda_3, \lambda_3)$ and, e.g., for $\lambda_3 = 1$,

$$-2\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}.$$

Note that this relation holds for all three matrices above. Why?

Augmented matrix; consistency

The **augmented matrix** of the linear system $A\mathbf{x} = \mathbf{b}$ is $A_{\mathbf{b}} = [A \ \mathbf{b}] \in \mathbb{R}^{m \times (n+1)}$.

Remark: Either $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ or $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A) + 1$.

Theorem (Rouch–Capelli): $A\mathbf{x} = \mathbf{b}$ is **consistent** $\iff \text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$
(equivalently $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$).

Moreover:

1. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = n$, the solution is **unique**. **Q:** What is it?
2. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) < n$, there are **infinitely many** solutions; choose $n - \text{rank}(A)$ variables **free**.

Solving with Gaussian elimination

Triangularization (Gaussian elimination)

Theorem

Applying elementary row operations to $A_{\mathbf{b}}$ transforms any system into an equivalent **upper-triangular** one.

This follows essentially from the fact that $A\mathbf{x} = \mathbf{b} \Leftrightarrow A_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0} \Leftrightarrow CA_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0}$ for any invertible C

$$\begin{array}{rcl} x_1 + x_2 + x_3 = 3 & & x_1 + x_2 + x_3 = 3 \\ x_1 - x_2 + 2x_3 = 2 & \iff & -2x_2 + x_3 = -1 \\ 4x_1 + 6x_2 - x_3 = 9 & & -4x_3 = -4 \end{array}$$

Row ops: $r_2 \leftarrow r_2 - r_1$, $r_3 \leftarrow r_3 - 4r_1$, $r_3 \leftarrow r_3 + r_2$.

We have $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 3 \Rightarrow$ unique solution $\mathbf{x} = (1, 1, 1)$.

A homogeneous example

Consider

$$\begin{aligned}a + b + c + d + e + f &= 0, \\2a + 2b + 2c + 2d - e - f &= 0, \\3a + 3b - c - d - e - f &= 0.\end{aligned}$$

Row-reducing the coefficient matrix gives

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix},$$

so there are 3 pivots, $\text{rank} = 3$, and the solution space has dimension $6 - \text{rank} = 3$:

$$(a, -a, c, -c, e, -e).$$

A basis is $\{(1, -1, 0, 0, 0, 0), (0, 0, 1, -1, 0, 0), (0, 0, 0, 0, 1, -1)\}$.

General solution structure

Theorem: Every solution of $A\mathbf{x} = \mathbf{b}$ can be written as

$$\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p,$$

where $\mathbf{x}_h$ is the **general solution** of $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_p$ is any **particular solution** of $A\mathbf{x} = \mathbf{b}$.

Corollary: If $A\mathbf{x} = \mathbf{b}$ is consistent, it has as many solutions as $A\mathbf{x} = \mathbf{0}$.

Example: Three identical equations

$$x_1 + x_2 + x_3 = 1$$

yield $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 1$. Choose two free variables:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}.$$

Two quick examples

1. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$ we found $\text{rank}(A) = 2$. Thus
 $\dim \ker(A) = 3 - 2 = 1$ and

$$\ker(A) = \{\lambda(-2, 1, 1) : \lambda \in \mathbb{R}\}.$$

2. Consider

$$\begin{aligned}x_1 + x_2 + x_3 &= -1, \\x_1 + 2x_2 + 4x_3 &= 2.\end{aligned}$$

Row reduction gives $x_1 + x_2 + x_3 = -1$, $x_2 + 3x_3 = 3$, so
 $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 2$. One free variable:

$$\{\lambda(2, -3, 1) + (-4, 3, 0) : \lambda \in \mathbb{R}\}.$$

Finding a basis for a span

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}.$$

Form the 3×3 matrix with these as **columns**:

$$\begin{pmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 0 & 3 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\text{rank} = 2$ and $3\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$. A convenient basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2/3 \end{pmatrix} \right\}$$

(or, equivalently, two of the original vectors that are independent).

Standing on the shoulders of giants



“If I have seen further, it is by standing on the shoulders of giants.” **Isaac Newton (1675)**.

Some of the giants of linear/matrix algebra:

- Gauss (Gaussian elimination, least squares, determinants)
- Cauchy (CauchySchwarz, eigenvalues, determinants)
- Laplace (Laplace expansion of determinants)
- Cayley (CayleyHamilton theorem, matrix theory)
- Sylvester (matrix rank, invariant theory)
- Jordan (Jordan canonical form)
- Frobenius (matrix factorizations, linear algebra foundations)

(pictured here as the base of a *castell*)

Chapter 8: Eigenvalues and Quadratic Forms

Eigenvalues/eigenvectors appear in growth models, Markov chains, PCA, and stability of dynamical systems.

Quadratic forms drive optimization via second-order conditions.

Read Chapter 10 of Werner–Sotskov (optional: Simon–Blume, Ch. 7).

Exercises 10.2; 10.5 A,B,C; 10.6(a,b).

Eigenvalues and eigenvectors: definitions

For $A \in \mathbb{R}^{n \times n}$, $\lambda \in \mathbb{R}$ is an **eigenvalue** if there exists $\mathbf{x} \neq \mathbf{0}$ with

$$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I_n)\mathbf{x} = \mathbf{0}.$$

Then $\mathbf{x}$ is an **eigenvector** for λ .

Characterization:

$$\lambda \text{ eigenvalue} \iff \text{rank}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

The **characteristic polynomial** $P(\lambda) = \det(A - \lambda I_n)$ has degree n ; the real roots (at most n) are the eigenvalues.

Procedure:

1. Solve $\det(A - \lambda I_n) = 0$ to get eigenvalues.
2. For each λ , solve $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ for eigenvectors.

Eigenvalues and eigenvectors: migration example

Consider a population split between two regions: city (x_t^C) and countryside (x_t^R) at time t . Suppose each year a fraction of people move between the two regions:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix} \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix} = A \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

Here, 0.8 means that 80% of city dwellers stay in the city, while 0.4 means that 40% of rural dwellers move to the city in the next period.

If the **urban/rural ratio stabilizes**, the next state must be a constant multiple of the current state:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = A \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix} = \lambda \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

The long-run stable distribution of the population is given by an **eigenvector** of A , and λ is the associated **eigenvalue** describing the growth rate.

With $A = \begin{pmatrix} 0.8 & 0.4 \\ 0.3 & 0.9 \end{pmatrix}$,

$$(0.8 - \lambda)(0.9 - \lambda) - 0.12 = 0 \iff \lambda^2 - 1.7\lambda + 0.6 = 0.$$

The solutions are $\lambda_1 = 1.2$ and $\lambda_2 = 0.5$.

For $\lambda_1 = 1.2$, eigenvectors are multiples of $(1, 1)$. For $\lambda_2 = 0.5$, eigenvectors are multiples of $(1, -\frac{3}{4})$.

For a population-growth interpretation we keep the dominant eigenpair: equal cohort proportions grow by 20% per step.

Basic spectral facts

If $A \in \mathbb{R}^{n \times n}$,

1. Eigenvectors associated with distinct eigenvalues are **linearly independent**.
2. If A is **symmetric**, then all eigenvalues are real and eigenvectors for distinct eigenvalues are **orthogonal** (Spectral Theorem).

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then $P(\lambda) = \lambda^2 - 2\lambda - 3$, so $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors along $(1, 1)$ and $(-1, 1)$, respectively.

Take the spectral decomposition $A = U\Lambda U^T$. Show columns of U are the eigenvectors.

Repeated eigenvalues

Consider

$$A = \begin{pmatrix} 0 & -1 & 1 \\ -7 & 0 & 5 \\ -5 & -2 & 5 \end{pmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 2)^2(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 2$ (algebraic multiplicity 2) and $\lambda = 1$ (algebraic multiplicity 1).

Eigenspaces:

$$E_2 = \ker(A - 2I) = \text{span}\{(1, -1, 1)\}, \quad E_1 = \ker(A - I) = \text{span}\{(2, 1, 3)\}.$$

Since $\dim E_2 = 1 < 2$, A is **defective** (not diagonalizable).

Application: Principal Component Analysis (PCA)

We observe n data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.

Step 1 (Centering): Arrange the data in a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows $\mathbf{x}_1, \dots, \mathbf{x}_n$. Define the centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and then the **centered data** is $H\mathbf{X}$.
(its rows are the centered data points $\mathbf{x}_i - \bar{\mathbf{x}}$)

Step 2 (Sample covariance):

$$S = \frac{1}{n}(H\mathbf{X})^\top(H\mathbf{X}).$$

Step 3 (Spectral decomposition):

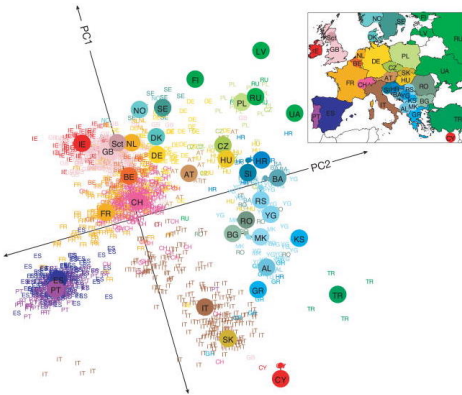
$$S = U\Lambda U^\top \quad \text{and define } \mathbf{Y} = H\mathbf{X}U \quad (\text{rotate the centered data}).$$

The sample covariance of $\mathbf{Y}$ is $\frac{1}{n}\mathbf{Y}^\top\mathbf{Y} = U^\top S U = \Lambda$ (diagonal! and so $\mathbf{y}_i$ are uncorrelated).

The columns of U are **principal directions** and diagonal entries of Λ are the **explained variances**.

Sorting eigenvalues in decreasing order ranks components by variance captured.

PCA: why it matters



Dimensionality reduction: compress high-dimensional data into a few directions that preserve most variability.

Noise filtering: small eigenvalues $\rightarrow$ directions with little signal.

Visualization: projecting onto first 2 PCs often reveals hidden structure.

Nice example: genetic variation in Europeans plotting individuals on the first two PCs of SNP data *recovers the geographic map of Europe!*

Numerical toy example

Covariance matrix:

$$S = \begin{pmatrix} 72.9 & 87.5 & 6.875 \\ 87.5 & 108.3 & 8.75 \\ 6.875 & 8.75 & 0.73 \end{pmatrix}.$$

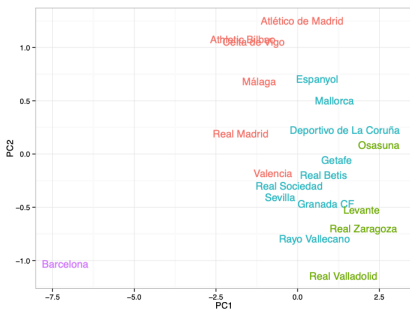
Spectral decomposition:

$$S = U\Lambda U^{\top}, \quad \Lambda = \text{diag}(180, 1.38, 0.005).$$

⇒ The first component explains > 99% of the variance; the last can be safely ignored.

PCA & Unique football styles

“Mille viae ducunt Barcinonem” (A thousand roads lead to Barcelona)



“Searching for a Unique Style in Soccer” (Gyarmati, Kwak & Rodriguez, 2014), studies sequential pass patterns in a team’s passing.

They defined “flow motifs” as statically significant pass subsequences (e.g., $A \rightarrow B \rightarrow C \rightarrow A$).

Applied motif analysis to compare styles across teams. The data are projected to the first two principal components.

Quadratic forms

A **quadratic form** is $Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i,j=1}^n a_{ij} x_i x_j$ with $A \in \mathbb{R}^{n \times n}$ symmetric.

Remark: Since the coefficient of $x_i x_j$ is $a_{ij} + a_{ji}$, every quadratic form has a unique symmetric representative (replace A by $\frac{1}{2}(A + A^\top)$).

Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 5/2 \\ 5/2 & 4 \end{pmatrix}$ define the same

$$Q(x, y) = x^2 + 5xy + 4y^2.$$

Signs of quadratic forms

For symmetric A :

1. Positive definite: $\mathbf{x}^\top A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
2. Negative definite: $\mathbf{x}^\top A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
3. Positive semidefinite: $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$.
4. Negative semidefinite: $\mathbf{x}^\top A \mathbf{x} \leq 0$ for all $\mathbf{x}$.
5. Indefinite: takes both positive and negative values.

Example:

$$\mathbf{x}^\top \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{x} = (x - y)^2 \geq 0$$

so the matrix is positive semidefinite.

Eigenvalue and minor tests

Spectral test (symmetric A):

- PD $\iff$ all eigenvalues > 0 ; ND $\iff$ all < 0 ;
- PSD $\iff$ all ≥ 0 ; NSD $\iff$ all ≤ 0 ;
- Indefinite $\iff$ eigenvalues of different signs.

Sylvester's criterion (leading principal minors D_k):

- PD $\iff D_k > 0$ for $k = 1, \dots, n$;
- ND $\iff (-1)^k D_k > 0$ for $k = 1, \dots, n$.

Example: $A = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$ has $D_1 = -3$, $D_2 = 5$, $D_3 = -25 \Rightarrow$
negative definite.

Examples

1. $A = \begin{pmatrix} 1 & 4 & 6 \\ 4 & 2 & 1 \\ 6 & 1 & 6 \end{pmatrix}$: $D_1 = 1$, $D_2 = -14$, $D_3 = -109 \Rightarrow$ indefinite.

2. $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$: $D_1 = 3$, $D_2 = 5$, $D_3 = 12 \Rightarrow$ positive definite.

3. The symmetric matrix of $Q(x, y, z) = xy + yz$ is indefinite since $D_1 = D_3 = 0$ and $D_2 = -\frac{1}{4}$.